arc_temp_humid - outstanding work

Updated 6 September 2026.

Everything described below is **not yet done**. What *is* done is in `CHANGELOG.md`; the architecture is in `CLAUDE.md`.

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1. Rename the GitHub repository - DONE

`actionresearchprojects/arc_tz_temp_humid` is now `arc_temp_humid`, and everything that depended on the old name has been dealt with.

The hazard here was that the main site's `sync-embedded.yml` derives its destination folder from the dispatch payload's `source_repo`. A rename alone would have started filling a new `embedded/arc_temp_humid/` while the published page went on iframing the old folder, frozen: no 404, no error, just a dashboard that silently stopped updating.

That was headed off by mapping **both** names to the same folder:

```
arc_tz_temp_humid|arc_temp_humid)    FOLDER="arc_temp_humid" ;;
```

so the repository and the site never had to change in lockstep. The live state:

URL	Status
`/graphs/arc-temp-humid/`	live, iframes `embedded/arc_temp_humid/`
`/graphs/arc-temp-humid/config`	live
`/explainers/arc-temp-humid/`	live, regenerated from the current README
`/graphs/arc-tz-temp-humid/`	redirects to the new address
`/explainers/arc-tz-temp-humid/`	redirects to the new address

The git remote, both workflow payloads and the header info-icon link now all use the new name.

One tidy-up left

`embedded/arc_tz_temp_humid/` is still present on the site repo: a dead copy of roughly 16 MB, superseded by `embedded/arc_temp_humid/`. It was kept as a fallback while the move settled and can be deleted whenever you like. Nothing references it.

Note the Pages URL for this repo changed with the rename, from `actionresearchprojects.net/arc_tz_temp_humid/` to `.../arc_temp_humid/`. Nothing links to it - the public route is `/graphs/arc-temp-humid/`.

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2. Automate the ARC UK Omnisense fetch

This is the only item still blocked on information I do not have.

The UK export is added **by hand** to `data/omnisense_uk/omnisense_uk_YYYYMMDD_HHMM.csv`. The timestamp is the fetch time and is what the sidebar's freshness note reads, so keep the format.

`fetch_omnisense.py` is already parameterised by site. To switch it on:

1. Fill in the commented-out `SITES["uk"]` entry, in particular **`site_nbr`**.
2. Add a `--site uk` step to `.github/workflows/update-dashboard-data.yml`, next to the existing Omnisense step, with the same `continue-on-error`.

Finding the site number

It is not in the CSV: the export carries site *names*, not ids, and the `download_935909450.csv` filename was a download-job id.

Log in at omnisense.com, select the site, and read `siteNbr=` from the address bar. That is where the Tanzanian `152865` came from - it appears in `fetch_omnisense.py` as `dnld_rqst.asp?siteNbr=152865`.

The site is named **"Simmonds.Mills Retrofits"** - the `site_name` on every block of the UK export.

Credentials are probably already correct

The UK export contains a sensor labelled `House 5 Metal Roof S.E` alongside the Grove, Holywell, Chestnuts and No. 59 sensors, so one login already sees both the Tanzanian and UK material. The existing `OMNISENSE_USERNAME` / `OMNISENSE_PASSWORD` secrets will very likely work, with only `site_nbr` differing.

If not, `SITES["uk"]` accepts `username_env` / `password_env` naming its own secrets, falling back to the shared pair when absent.

When it runs

The UK export is one CSV covering several sites - Grove, Holywell, Chestnuts, No. 59 - of which this dashboard uses four sensors. `GROVE_SENSORS` and `HOLYWELL_SENSORS` in `build.py` filter by sensor id, so the rest are ignored rather than plotted. Those two sets are what to update if sensors change.

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3. Decide the UK overheating threshold

Currently **none**: the three UK datasets have `"threshold": None`, so no red band is drawn and the sidebar control is hidden rather than offering a meaningless toggle. Tanzania keeps 32-35 C.

No UK figure was assumed because the sensible source, CIBSE TM52 / TM59, is **per room type**. TM59 uses 26 C for bedrooms, but Grove's sensor is a living room and Holywell's is "Internal Ambient".

Once decided it is one line per dataset:

```
"threshold": {"low": 26, "high": 28},
```

The label, the band on all three chart types and the sidebar control follow automatically.

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4. Confirm the UK Open-Meteo coordinates

Done and live, but with one assumption worth correcting.

Both UK buildings now have their own Open-Meteo feed. Each drives its own building's adaptive comfort running mean and both appear on the ARC UK chart. They are fetched daily alongside the Tanzanian feed.

The coordinates in `fetch_openmeteo.py` are **town-centre positions** for Hereford and Criccieth, not the buildings themselves:

```
"grove":    {"lat": 52.0567, "lon": -2.7160, ...}   # Hereford
"holywell": {"lat": 52.9186, "lon": -4.2372, ...}   # Criccieth
```

Open-Meteo's historical model runs on a roughly 11 km grid, so a town-centre fix is usually the same grid cell as the building - but if the exact positions (or postcodes) are to hand, replace the two lat/lon pairs. It is a one-line change per site and needs no other edits; the next daily run picks it up.

As a sanity check that the two feeds are genuinely distinct: Grove averages 1.03 C warmer than Holywell across 30,497 paired hours, with only 611 hours identical - consistent with inland Herefordshire against coastal North Wales.

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5. Known limitation, not a task

ASHRAE 55 Section 5.4.1(a) requires that no heating is in operation. Nothing at any site records heating status, and nothing in a temperature and humidity trace reliably separates a heated room from a merely warm one.

The dashboard states this rather than inferring it: a bolded line in the sidebar applicability panel directly beneath the quoted criterion, and in the tooltips explaining the method. The 10-33.5 C running-mean gate is enforced automatically and greys out readings the standard does not cover.

That gate now has teeth it did not have before, because the UK Open-Meteo feeds reach back to March 2023: once UK winter indoor data exists, cold-weather readings will be greyed and excluded from the comfort percentage. But **a mild day with the heating on will still pass every test the dashboard can apply**.

Closing this properly needs a heating-status input, not a cleverer algorithm. See `adaptivecomfort.md` section 2 for the full reasoning, including why calendar-month filtering was considered and rejected.